

Alexander DeLise

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EDUCATION

Florida State University

May 2027

B.S. Applied and Computational Mathematics, B.S. Computational Science, GPA - 4.0

Tallahassee, FL

- **Awards:** Presidential Scholar, Florida Bright Futures Scholar, Pi Mu Epsilon Mathematics Honor Society
- **Relevant Coursework:** Computational Probabilistic Modeling, Data Mining and Machine Learning, High Performance Computing, MCMC Methods, Discrete/Continuous Algorithms for Science, Math Modeling, Statistics
- **Honors in the Major Thesis:** Generative AI and Compressed Sensing, advised by Dr. Nick Dexter

INDUSTRY EXPERIENCE

Johns Hopkins University Applied Physics Laboratory

May 2026 – August 2026

Incoming Data Science Intern

Laurel, MD

- KBS Systems Performance Analysis Group

RESEARCH EXPERIENCE

Florida State University

Nov 2025 – Present

Machine Learning Researcher – Generative Models & Inverse Problems

Tallahassee, FL

- Achieved $\sim 2\times$ higher reconstruction quality (PSNR, SSIM) than baseline heuristics in extremely low-sampling compressed-sensing regimes using diffusion-based generative priors
- Developed end-to-end reconstruction pipelines and running large-scale evaluations on natural-image datasets to validate performance and robustness across sampling patterns and noise levels
- Implemented data pipelines and model training/inference workflows in Python using HuggingFace and PyTorch

Emory University

June 2025 – August 2025

Research Fellow – Scientific Machine Learning | 🔗

Atlanta, GA

- Developed mathematical, closed-form baselines for optimal rank-limited encoder-decoder models, providing a principled reference for what learned autoencoders should achieve on forward and inverse problems
- Evaluated these baselines across a variety of experimental settings to demonstrate generalizability: biomedical imaging deblurring, financial factor analysis (194 stocks over 787 trading days), and fluid dynamics (shallow water equation inversion)
- Implemented and trained encoder-decoder models in PyTorch across large datasets with hyperparameter tuning; demonstrated that mathematical baselines consistently outperform the trained models under the same rank constraint
- Collaborated in a diverse team, honing communication, adaptability, and joint problem-solving to drive research progress
- First author on manuscript on scientific machine learning and neural networks, under submission to a peer-reviewed journal

University of Tennessee, Knoxville

May 2024 – April 2025

Research Fellow – Quantum Algorithms | 🔗

Knoxville, TN

- Built an end-to-end Python preprocessing pipeline to generate, validate, and package feasible solutions for constrained binary optimization problems for downstream experimentation with quantum computers
- Designed a scalable state-labeling approach using auxiliary variables, improving solution accuracy by up to $360\times$ versus baseline heuristics
- Implemented and benchmarked optimization workflows in Python, integrating custom state preparation and running controlled comparisons across algorithm variants
- Created a reproducible evaluation suite by serializing experiment configurations into callable files, enabling automated sweeps and consistent reruns for reproducible results
- Collaborated in a cross-disciplinary research team to debug implementations, share findings, and deliver steady progress
- Co-authored a research manuscript describing the method and experimental results (under review)

FAMU-FSU College of Engineering

September 2023 – June 2025

Undergraduate Researcher – Operations Research | 🔗

Tallahassee, FL

- Designed/implemented large-scale optimization models for pandemic-era patient allocation and healthcare facility planning
- Processed and analyzed extensive COVID-19 datasets (hospitalization, vaccination, geospatial) and integrated them with epidemic vaccine dynamics and policy constraints into a decision-making framework
- Simulated 1 million+ patient transfers, achieving a 16.4% reduction in unmet hospital demand, identifying bottlenecks
- Developed pipelines in Python, R, CPLEX, and GAMS for data processing, model implementation, and data visualization
- First author on a paper accepted to the 2025 North American IEOM Conference in Orlando, FL

TECHNICAL SKILLS AND INVOLVEMENTS

Languages & Tools: Python, C++, Git, GitHub, VS Code, Gurobi (gurobipy), IBM ILOG CPLEX, GAMS, Microsoft Office

Frameworks & Libraries: PyTorch, HuggingFace, scikit-learn, NumPy, pandas, SciPy, PennyLane

Involvements: JPMorganChase Data for Good Hackathon (Participant); Pi Mu Epsilon Mathematics Honor Society (President); Society for Industrial and Applied Mathematics (SIAM)

Interests: Weightlifting, pickleball, and all Tampa Bay sports teams